

Joseph Milliken

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EDUCATION

Bachelor of Science in Computer Science and Engineering, *University Of Colorado Denver* 08/2022 – 05/2026 | Denver, CO
GPA: 3.94/4.0

PROFESSIONAL EXPERIENCE

Teaching Assistant | Algorithms & Principles of Programming Languages 08/2024 – 05/2026 | Denver, CO
University of Colorado Denver

- Assisted in teaching undergraduate courses in algorithms and programming languages, reinforcing concepts such as **computational complexity**, dynamic programming, greedy algorithms, **language semantics**, and type systems.
- Led regular office hours and provided individualized academic support, helping students improve comprehension, and problem solving skills.

PROJECTS

Clara | AI-Powered Pathology Insights, *MERN, Python, AWS S3*

- Engineered an end-to-end **AI platform** for whole-slide image (WSI) biopsy classification, leveraging **PyTorch**, **FastAPI**, **Docker**, and **Amazon S3** to support pathology workflows.
- Applied **Agile** software development practices within a four person team over two semesters to deliver a clinical decision platform that reduced pathology second-opinion turnaround times from up to **two weeks** to approximately **five seconds**.
- Earned **1st Place** among Computer Science teams at the College of Engineering Senior Capstone Expo for developing an AI-powered clinical decision-support platform with healthcare impact.

Transferr | Peer-2-Peer Networking Tool, *C#, Socket Programming, Avalonia UI*

- Engineered a cross-platform peer-to-peer file transfer application using **C#**, **.NET**, and **Avalonia**, enabling direct file transfers over TCP/IP across Windows, macOS, and Linux.
- Designed a custom transfer protocol to transmit **JSON metadata and raw file data in 80 KB chunks**, with real-time progress and transfer status reporting.
- Implemented **SHA-256 checksum verification** to validate file integrity after transmission and detect corrupted or incomplete transfers.

taskctl | Task Manager, *Golang, JSON*

- Developed a lightweight, **cross-platform** command-line task manager.
- Implemented core task operations: add, update, delete, mark complete/incomplete, and clear all tasks.
- Enhanced usability with features like priority and due dates, sorting by status/priority/date.
- Designed a polished interactive UI with promptui, supporting intuitive keyboard navigation.

SKILLS

Languages: Python, C/C++, Golang, C#, SQL, Javascript/Typescript, CSS, HTML, Bash

Developer Tools: Git/Github, Copilot, Render, MongoDB, Visual Studio/VS Code, Arduino IDE, phpMyAdmin, Postman, ChatGPT

Frameworks and Libraries: Express.js, React.js, pandas, numpy, matplotlib

Platforms: Docker, macOS, Windows, AWS (S3), PostgreSQL

CERTIFICATES

NVIDIA: Applications of AI for Anomaly Detection

Docker: Foundations Professional Certification

Atlassian: Agile Project Management Professional Certificate

CU Denver: Cybersecurity and Secure Computing

AWARDS

Dean's List

Fall 22 | Spring 23 | Fall 23 | Spring 24 | Fall 24 | Spring 25 | Fall 25 | Spring 26

LANGUAGES

Spanish — Native/Bilingual